Drift-Regularized Multi-Token Prediction via On-Policy Self-Distillation

Po-Wei Huang, Feng-Ting Liao
Department of Physics, University of Oxford
MediaTek Research

1 objective

Standard next-token prediction generates one token per step, incurring high memory-bandwidth overhead and inference latency during long-form generation and reasoning-intensive tasks. MTP mitigates this bottleneck by generating a contiguous block of k tokens in a single forward pass. We build on on-policy self-distillation, in which a student proposes future tokens at inserted mask positions and a frozen teacher supplies supervision conditioned on the student-generated continuation [7].

Our objective is to reduce the capability loss that accompanies longer parallel prediction horizons. We will test the hypothesis that, at comparable realized acceleration, latent regularization changes both the geometry of predictive states and the quality–acceleration trade-off, with potentially different outcomes for pointwise regression and population-level alignment. Success will be assessed through task scores relative to the original sequential teacher, confidence calibration, held-out feature-distribution diagnostics, and measured throughput and latency. This ordering is a hypothesis to be tested rather than an established result.

2 novelty

A correct token prediction constrains relative logit margins without uniquely determining the hidden representation. Consequently, newly inserted MTP mask states must learn parallel future-token computations without explicit supervision over their representation geometry. Work on feature-based speculative decoding, including EAGLE [8], motivates addressing feature uncertainty; NextLat [9] also provides motivation for auxiliary prediction in latent space.

Paired SmoothL1 regression supplies dense supervision but imposes a one-to-one Euclidean match between each parallel student state and its sequential teacher counterpart. Our proposed contribution is to adapt population-level drifting [3] to conditional MTP representation alignment. A local vector field attracts student states toward teacher feature neighborhoods and repels them from crowded student neighborhoods. We use a feature-distance kernel without a prefix-similarity factor; any horizon-specific comparisons are implemented through the selection of feature pools rather than through a context kernel. We will compare horizon-wise alignment with block-level alignment to determine whether modeling inter-token feature dependencies improves chunk coherence.

3 impact

If successful, this approach could extend the usable parallel prediction horizon while retaining more of the sequential model's reasoning capability. The regularizer is applied during training and adds no inference module to the standalone MTP model. The intended practical benefit is improved generation quality at a given latency or greater throughput at a given quality target. These benefits require system measurements and cannot be inferred from the configured block size alone.

Preprint.

We compare conventional token-level MTP distillation with two auxiliary objectives: SmoothL1 regression, which directly aligns paired student and teacher hidden states, and drifting regularization, which uses attraction and repulsion across feature populations. This separates pointwise matching from distributional alignment. Calibration and task accuracy will be reported independently of feature-distribution distances. A later extension could apply the same conditional regularizer to a speculative drafter such as DFlash [1]; the initial study will focus on standalone MTP.

4 Methods

4.1 On-policy MTP self-distillation

For training region r, let c_r denote a context prefix and k_r the number of future predictions. Following Kirchenbauer et al. [7], the flattened student input inserts $k_r - 1$ trainable <MTP> mask tokens. Repeated position IDs and a blocked causal attention mask allow active prediction states to attend to the prefix and preceding masks without altering copied context representations. At horizon j, the student produces hidden state $u^S_{r,j}$ and token distribution $p^S_{r,j}$. Its greedy proposal is $\hat{y}_{r,j} = \arg\max_v p^S_{r,j}(v)$.

The frozen teacher is evaluated on the student-forced continuation:

$$u^T_{r,j} = H_{\theta_T}(c_r, \hat{y}_{r,<j}), \qquad \bar{y}_{r,j} = \arg\max_v p_{\theta_T}(v \mid c_r, \hat{y}_{r,<j}). \tag{1}$$

With $N_{\text{act}} = \sum_r k_r$, the hard on-policy distillation loss is

$$\mathcal{L}_{\text{MTP}} = -\frac{1}{N_{\text{act}}} \sum_{r,j} \log p^S_{r,j}(\bar{y}_{r,j}). \tag{2}$$

For active index $i = (r, j)$, let $x_i = \phi_S(u^S_i)$ and $y^+_i = \phi_T(u^T_i)$ be features in a shared latent space, using the same normalization. Detached student features $y^-_l = \text{sg}(x_l)$ form the negative population, where sg denotes stop-gradient.

4.2 Latent regularization

We compare the unified objective

$$\mathcal{L} = \mathcal{L}_{\text{MTP}} + \lambda_R \mathcal{L}_R, \qquad R \in \{\text{SmoothL1}, \text{drift}\}. \tag{3}$$

The paired regression baseline uses SmoothL1 (Huber-type) regression:

$$\mathcal{L}_{\text{SmoothL1}} = \frac{1}{N_{\text{act}} d} \sum_{i=1}^{N_{\text{act}}} \sum_{m=1}^{d} \rho_\beta(x_{i,m} - \text{sg}(y^+_{i,m})), \tag{4}$$

where $\rho_\beta(z) = z^2/(2\beta)$ for $|z| < \beta$ and $\rho_\beta(z) = |z| - \beta/2$ otherwise. The implementation's β value will be reported with the final experimental configuration. For drifting, the feature-only Laplace kernel is

$$\kappa_h(x_i, y_l) = \exp\left(-\frac{\|x_i - y_l\|_2}{\tau_h}\right). \tag{5}$$

Positive and negative pools $\mathcal{I}^+_i$ and $\mathcal{I}^-_i$ contain teacher and detached student features, respectively. No prefix similarity term is used. Pool selection and any horizon grouping are separate from the kernel definition. For $s \in \{+, -\}$, define $\mathcal{J}^+_i = \mathcal{I}^+_i$ and $\mathcal{J}^-_i = \mathcal{I}^-_i \setminus \{i\}$. Normalize affinities as

$$a^s_{il} = \frac{\kappa_h(x_i, y^s_l)}{\sum_{m \in \mathcal{J}^s_i} \kappa_{\text{cond}}(i, m; y^s_m) + \epsilon}, \qquad l \in \mathcal{J}^s_i, \tag{6}$$

with $\epsilon > 0$ for numerical stability. The self-excluded drift estimate is

$$\widehat{V}_i = \sum_{l \in \mathcal{J}^+_i} a^+_{il}(y^+_l - x_i) - \sum_{l \in \mathcal{J}^-_i} a^-_{il}(y^-_l - x_i). \tag{7}$$

Table 1: Planned objective variants and their diagnostic roles.

ID	Objective	Diagnostic role
M0	Hard on-policy MTP	Token-only baseline
M1	M0 + paired latent SmoothL1	Pointwise alignment
M2	M0 + conditional full drift	Attraction and repulsion
M3	M0 + attraction-only drift	Repulsion ablation
M4	M0 + block-level full drift	Joint block coherence
M5	Soft on-policy KL distillation	Soft token supervision

Following the stopped-target construction of drifting models [3], the auxiliary objective is

$$\mathcal{L}_{\text{drift}} = \frac{1}{N_{\text{act}}} \sum_i \left\| x_i - \text{sg}\left(x_i + \eta \frac{\widehat{V}_i}{\|\widehat{V}_i\|_2 + \epsilon} \right) \right\|_2^2, \tag{8}$$

where η controls the target displacement. The block-level variant applies the same construction to concatenated normalized block features, $b_r^S = k_r^{-1/2} \,\text{vec}[x_{r,1}; \ldots; x_{r,k_r}]$, and their teacher counterparts, grouping blocks of the same length.

4.3 Controlled comparisons

The planned variants are summarized in Table 1. We will keep initialization, data, and training budgets comparable.The current strategy is to fix gradient-norm ratio between regularisation part and original MTP one to be approximately 0.30. All the studies below are freezed under this ratio to test under 1B Llama-3.1-Instruct model.

4.4 Evaluation and decision rules

Current 1B experiments train on MetaMathQA and evaluate on GSM8K using free-rollout generation. Additional benchmarks for larger models will be selected in a later phase, after establishing robust results on the present setup. We currently report GSM8K answer accuracy and the realized average number of emitted tokens per forward pass. The frozen teacher under sequential decoding ($k = 1$) provides the capability-retention reference. All model-selection decisions will use held-out validation data.

We will measure token-level expected calibration error (ECE), following the calibration perspective of Joshi et al. [6]. Separately, we will compare teacher and student feature populations using sliced Wasserstein-2, maximum mean discrepancy [5], Sinkhorn transport cost [2], and 1-nearest-neighbor two-sample accuracy, with complementary distributional diagnostics motivated by Esteban-Casadevall et al. [4]. Features will use shared normalization and any PCA projection fitted only on training features. Diagnostics will be reported per horizon; nearest-neighbor accuracy approaching 50% indicates reduced population distinguishability under balanced sampling.

We will evaluate both static and confidence-adaptive decoding. Realized token acceleration is

$$k_{\text{eff}} = \frac{\text{emitted non-padding tokens}}{\text{decoding forward passes}}, \tag{9}$$

which will be reported alongside measured throughput and end-to-end latency. For a single operating-point summary, validation will select the fastest setting within one percentage point of the sequential teacher's validation accuracy, when such a setting exists; it will then be frozen for test evaluation. We will report uncertainty over evaluation examples and, where feasible, training seeds. If SmoothL1 outperforms drift, or token-only training outperforms both regularizers, we will report that outcome and examine sampling noise, regularization strength, and conditioning rather than assume the proposed ordering holds.

5 Risks

Noisy or mismatched feature populations. Conditional drift depends on the size and relevance of local teacher and student pools. Sparse horizon groups or unrelated contexts may produce noisy

updates. We will monitor pool sizes, kernel affinities, feature norms, and per-horizon diagnostics; cap or subsample pools for tractability; and retain paired SmoothL1 and attraction-only drift as diagnostic alternatives. Updates with insufficient pool support will be skipped.

Overregularization and misleading alignment. Forcing feature alignment may reduce token accuracy or encourage feature collapse. Matched auxiliary gradient strength and coefficient sweeps will help separate the effect of the regularizer from loss scaling. Lower latent distances or ECE alone will not be treated as evidence of improved reasoning: task scores and the teacher-retention gap remain necessary checks.

Training overhead and deployment trade-offs. Pairwise kernel interactions increase training compute and memory, and a larger configured block size need not yield proportional wall-clock speedup. We will bound population sizes and profile teacher passes, kernel computation, and training memory. Deployment comparisons will use identical hardware, batch sizes, prompt lengths, and generation limits.

Generalization and evaluation validity. Training on mathematical traces may degrade other capabilities, motivating the broader evaluation suite. Validation and test splits will remain separate, and training data will be checked for benchmark overlap. Faster generation may also amplify existing model errors; capability retention and confidence calibration will be assessed before making practical deployment claims.

6 Timeline & Resources

The project has completed baseline reproduction and is currently transitioning from drifting pilot experiments to systematic ablations. Dates and durations for the remaining phases depend on compute availability.

1. **Baseline reproduction (completed):** establish the on-policy MTP pipeline, sequential teacher reference, paired SmoothL1 baseline, and evaluation harness.
2. **Drift implementation and 1B pilot (in progress):** implement conditional pools, self-exclusion, and stopped-target updates; profile memory and runtime; Adjusting regularisation strength will be operated under the actions of running 8B Llama model. Under this intermediate step, proving drifting effectively gains advantages over baselines is the main objective.
3. **Main experiments and ablations (next):** run the block-size and confidence-threshold sweeps, compare all objective variants, and collect task, calibration, and latent-alignment measurements.
4. **Analysis and reporting (planned):** quantify uncertainty, analyze failure cases and capability retention, and prepare reproducible configurations and the final manuscript.

Required resources include a pretrained autoregressive checkpoint, a frozen teacher copy, GPUs with sufficient memory for student training and teacher forward passes, checkpoint storage, and held-out evaluation data. A pilot run will determine feasible batch and pool sizes and measure GPU-hours per run; the total budget will then be estimated from the planned configurations and seeds. The checkpoint, GPU type and count, storage budget, personnel allocation, and phase durations are **TBD**.

6.1 Preliminary Results and Current Progress (October 2026)

The on-policy MTP baseline has been reproduced, and pilot comparisons of unregularized MTP, SmoothL1, drifting, and combined objectives have been performed at the 1B scale. The following observations are preliminary; plots summarize the supplied experimental runs rather than a completed multi-seed significance study. The experimental configurations are summarized in Table 2. We compare hard-teacher MTP self-distillation against drifting regularization, paired SmoothL1 feature regression, and their combination. Additional ablations investigate kernel bandwidth selection and the maximum training prediction horizon $k_{\max}$.

Table 2: Experimental configurations and corresponding figure legends. Auxiliary weights (w) are loss coefficients, while percentages indicate approximate auxiliary-to-MTP gradient-norm ratios. All experiments use hard-teacher on-policy MTP cross-entropy as the primary training objective.

Figure legend	**Training configuration**
`nodrift`	MTP loss only, using hard-teacher cross-entropy supervision.
`drift selfpos w0.05`	MTP loss with self-positive drifting regularization ($w_{\text{drift}} = 0.05$), corresponding to an approximately 30% auxiliary-to-MTP gradient-norm ratio.
`drift selfpos w0.05 tau0.02-0.1-0.2`	MTP loss with self-positive drifting regularization ($w_{\text{drift}} = 0.05$; approximately 30% gradient-norm ratio) and adjusted kernel bandwidths ($\tau = 0.02,\ 0.1,\ 0.2$).
`nodrift kmax5`	MTP loss only, with the maximum training prediction horizon restricted to $k_{\max} = 5$.
`smoothl1 w0.53`	MTP loss with paired SmoothL1 feature regression ($w_{\text{SmoothL1}} = 0.53$), corresponding to an approximately 20% auxiliary-to-MTP gradient-norm ratio.
`drift selfpos w0.05 + smoothl1 w0.53`	MTP loss with both self-positive drifting ($w_{\text{drift}} = 0.05$; approximately 30%) and paired SmoothL1 regression ($w_{\text{SmoothL1}} = 0.53$; approximately 20%).

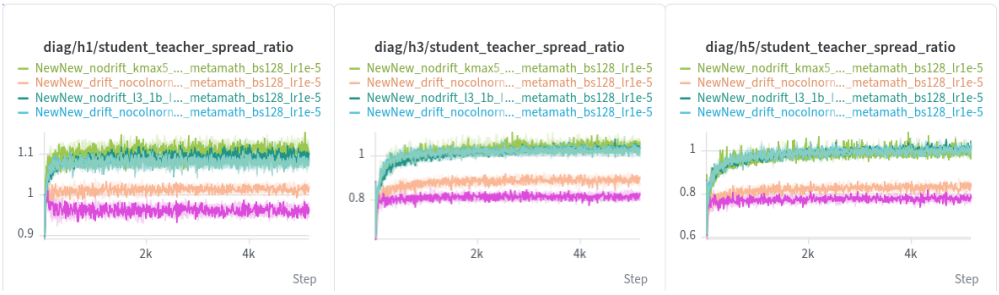


Figure 1: Student-to-teacher hidden-feature spread ratios over training steps for horizons 1, 3, and 5. Values near one indicate similar measured spread, not necessarily identical feature distributions. Run names are preserved from the experiment dashboard.

Feature dispersion across horizons. Figure 1 reports the student-to-teacher feature spread ratio, defined as the mean pairwise distance between distinct student hidden states divided by the corresponding mean distance between teacher states. A ratio below unity indicates a more contracted student distribution, while a ratio above unity indicates greater dispersion. SmoothL1 systematically contracts student representations, with the effect becoming more pronounced at longer prediction horizons. In contrast, drifting better preserves the teacher's feature spread, suggesting that population-level alignment can maintain relative geometric structure without forcing individual student states to collapse toward their paired teacher targets.

Distributional diagnostics. Figure 2 compares MMD and Sinkhorn distance across prediction horizons. Drifting configurations generally yield lower values than the unregularized baseline in the plotted range. SmoothL1 is competitive and lower than drifting in terms of Sinkhorn distance, as the pair distance between student and teacher is the goal for minimisation under smoothL1 training purpose, which is the quantity optimal transport values the most. MMD which cares about about two individual spread of groups will capture how alignment two generated distributions are. The results shows that drifting chemem has better alignments across MTP region than smoothL1 one, whereas under NTP(k=1) region smoothL1 still performs the best. Both metrics generally worsen as the horizon increases.

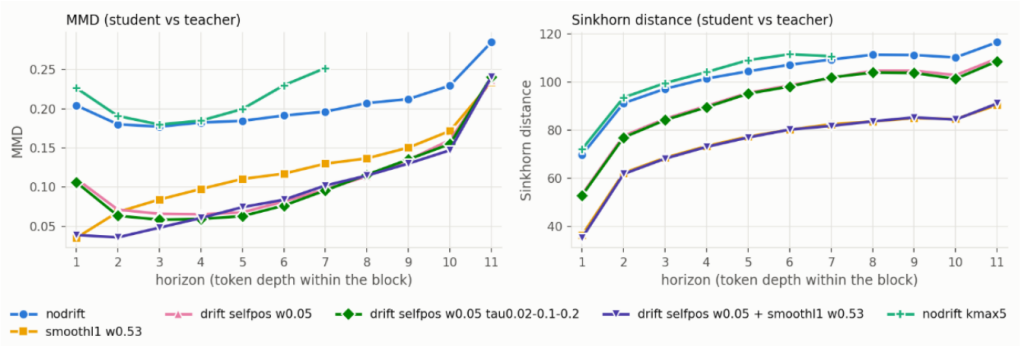


Figure 2: Held-out student–teacher maximum mean discrepancy (left) and Sinkhorn distance (right) as a function of prediction horizon. Lower is better for both diagnostics, but neither directly measures task accuracy. The different objectives exhibit horizon-dependent trade-offs.

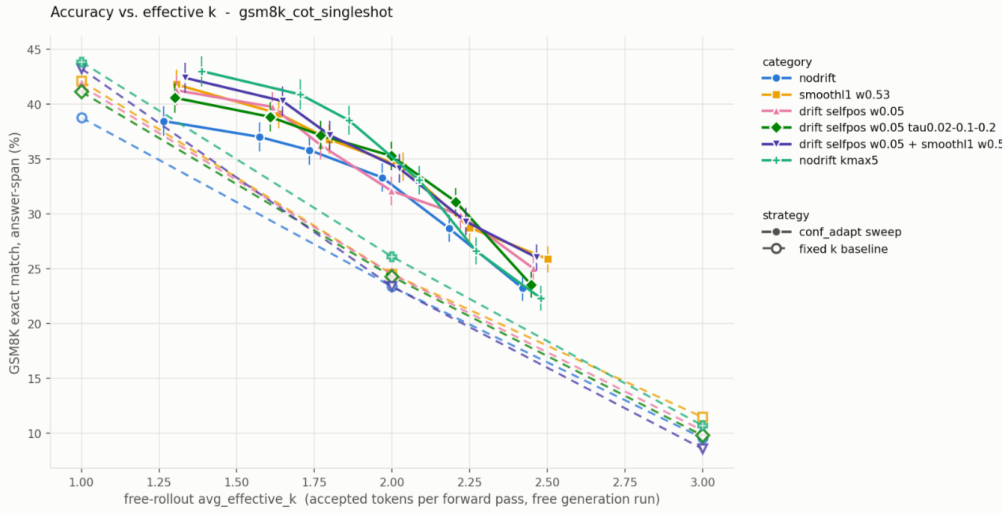


Figure 3: GSM8K free-rollout answer accuracy versus realized effective k for the 1B pilot runs. Solid lines show confidence-adaptive sweeps; dashed lines show fixed-k results. Higher accuracy and larger effective k are preferred, and comparison should be made at matched operating points rather than by configured $k_{\max}$ alone.

Free-rollout accuracy and acceleration. Figure 3 reports GSM8K single-shot chain-of-thought accuracy against the average effective k from free generation. Solid curves represent confidence-adaptive decoding and dashed curves fixed-k settings. At some matched acceleration values, SmoothL1 and drifting runs outperform the unregularized model, but no single objective dominates throughout the displayed range. The reduced-$k_{\max}$ run shows encouraging accuracy in part of the adaptive-decoding regime, although this does not establish a uniform gain from decreasing $k_{\max}$.

Next steps. We will test the stability of these observations under matched training budgets and repeated seeds, quantify paired feature errors separately from distribution distances, and examine whether horizon-specific feature geometry predicts improvements in free-rollout accuracy. We will also study the effect of $k_{\max}$ and expand to larger checkpoints and broader benchmarks once the 1B methodology is consolidated.

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